

A Peek into the Consensus Mechanisms of Open Source Blockchains



Explore three different consensus algorithms, comparing their tenets, advantages, and disadvantages, while also going into how they have influenced the development of open source blockchain ecosystems.

Consensus algorithms allow blockchain networks to operate safely and reliably. In a decentralised network, these algorithms guarantee that all users concur on the blockchain's present state. Three consensus mechanisms have become popular in the context of open source blockchains: Byzantine Fault Tolerance (BFT), Proof of Work (PoW), and Proof of Stake (PoS).

Consensus in blockchains

The protocols that allow blockchain networks to agree on a single data

value or version of the blockchain are known as consensus algorithms. The decentralised ledger's continuity, security, and integrity are all vitally dependent on them. The decentralised nature of blockchain would not be feasible without consensus, potentially resulting in vulnerabilities and inconsistent data. Data is updated and maintained by a single person (such as a database administrator) in conventional centralised systems. Decentralised systems, on the other hand, do not need a central authority because they rely on consensus processes to guarantee that

all participants agree on the authenticity of the data.

Decentralisation, security, fault tolerance and consistencies are vital characteristics of consensus algorithms.

Decentralisation: Here, a central authority is not required to verify transactions. Authority, control, and data are dispersed throughout a network as opposed to being centralised in one organisation or person. Decentralisation eliminates single point of failures..

Multiple autonomous nodes take part in a decentralised consensus method, which makes it resistant to